Rational expressions

1

a $\frac{7v}{5}$ **b** $\frac{3}{4}$ **c** $\frac{7v}{71}$ **d** $\frac{7x}{3}$

e $2p^4$ **f** $\frac{10n}{3m}$ **g** $\frac{-2k}{3}$ **h** $-\frac{y}{4x}$

2

a

i $\frac{4a}{4a(b-5c)}$ **ii** $=4a$ **iii** $\frac{1}{b-5c}$

b

i $\frac{2f(g-3h)}{2f}$ **ii** $2f$ **iii** $g-3h$

c

i $\frac{3r(2+st)}{3r(s-4t)}$ **ii** $3r$ **iii** $\frac{2+st}{s-4t}$

d

i $\frac{4xy(1+2y)}{12x^3y^3}$ **ii** $4xy$ **iii** $\frac{1+2y}{3x^2y^2}$

3

a $\frac{3x-8}{6}$ **b** -1 **c** -3 **d** $\frac{x-4}{3x+5}$

e $v-4w$ **f** $\frac{1}{v-5w}$ **g** 2 **h** 3

4

a $-(a+9)$ **b** $\frac{1}{x+5}$ **c** $\frac{1}{x+6}$ **d** $m-1$

e $m+10$ **f** $\frac{5m+7n}{8m}$ **g** $\frac{1}{x-1}$ **h** $\frac{1}{x+6}$

i $\frac{6}{5(x+12)}$ **j** $x+5$ **k** $\frac{x+2}{10}$ **l** $\frac{2(x+11)}{5}$

5

a $\frac{2}{r+4}$ **b** $\frac{2}{x}$ **c** $-\frac{5}{2x}$ **d** $\frac{5}{x^2}$

e $\frac{4}{r-5}$ **f** $\frac{p-q}{p+q}$ **g** $-\frac{3}{x}$ **h** $\frac{1}{3}$

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6

a $\dfrac{x+4}{x-5}$ **b** $\dfrac{x-7}{x-8}$ **c** $\dfrac{2x+3}{4x-5}$ **d** $\dfrac{3x+2}{6x-5}$

e $\dfrac{x-7}{x+7}$ **f** $\dfrac{x+4}{x}$ **g** $\dfrac{a+6}{a-4}$ **h** $\dfrac{p+20}{p-11}$

i $\dfrac{2x+3}{3-2x}$ **j** $-\dfrac{4x+5}{2+x}$ **k** $-\dfrac{x+4}{2(2+x)}$ **l** $\dfrac{x+2}{x(x+4)}$

7

a $x-5$ **b** $\dfrac{9x}{10(x+1)}$ **c** $\dfrac{x+7}{x-3}$ **d** $\dfrac{x-5}{x+10}$

e $\dfrac{4(x+4)}{x+5}$ **f** $\dfrac{x(x-5)}{x+3}$ **g** $-\dfrac{5+4x}{x+3}$ **h** $\dfrac{x-3}{5x-2}$

i $\dfrac{2(4x-5)}{4x+5}$ **j** $\dfrac{2x-1}{4x-5}$ **k** $\dfrac{x-5}{x+1}$ **l** $\dfrac{4x-1}{3-x}$

8 Luciana made an error in Step 2. She cannot divide out $2x$ because it is not a factor of the numerator. In fact, the result of Step 1 is already fully simplified.

9 Qin made an error in Step 3. The additive property of fractions is $\dfrac{A+B}{C}=\dfrac{A}{C}+\dfrac{B}{C}$, but instead Qin tried to use $\dfrac{A}{B+C}=\dfrac{A}{B}+\dfrac{A}{C}$ which is not true. In fact, the result of Step 2 is already fully simplified.

10

a k can be any multiple of 4. **b** $k=13$

c $k=0$ **d** $k=2$ and $k=-9$

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